

ARB Basket Fair Value

Regime-Gated Indicator Tutorial & Whitepaper

How to read each overlay, tune the settings, and identify higher-quality ARB+ / ARB- signals

Internal Use Notice

This document is an educational and operational guide for the ARB Basket Fair Value indicator. It is not financial advice, not a trading recommendation, and not a guarantee of performance. Use the framework for research, monitoring, and disciplined trade review.

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1. Executive Summary

The ARB Basket Fair Value indicator compares the chart symbol against a synthetic fair value built from a weighted peer basket. The goal is not to predict direction in isolation. The goal is to identify when the chart symbol is rich or cheap versus the basket, then wait for evidence that the relative-value spread is actually starting to mean revert.

The original indicator already created a synthetic fair value line, spread points, spread percent, z-score, ARB+ / ARB- labels, and a summary table. The upgraded regime-gated version changes the role of the signal. A large spread is no longer treated as a trade entry. It is treated as Step 1 only: a dislocation alert.

Core upgrade

ARB+ should print only after five checks pass: dislocation exists, the basket relationship is valid, squeeze/event risk is clear, residual spread is compressing, and signal cooldown allows a fresh print. This reduces repeated false positives in names that are moving because of forced flow, news, squeeze/unwind, or broken correlation.

Old interpretation	New interpretation
Price below synthetic fair value = ARB+	Price below synthetic fair value = D+ dislocation only.
Signal can print while spread keeps widening.	Signal waits for residual compression.
Basket validity assumed.	Correlation, R2, and beta must pass.
Event/squeeze regimes not filtered.	RVOL, ATR, gap, z-jump, and extreme residual filters can block signals.
Labels can cluster during one-way selloffs.	Cooldown plus event gate reduces clustered false positives.

2. Core Concept

2.1 Synthetic Fair Value

Synthetic fair value is an estimate of where the chart symbol might be trading if it followed the weighted return of the selected basket. It is not intrinsic value. It is a relative-value anchor based on peer movement.

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basket return = weighted average of active peer returns
chart return  = chart close / chart anchor - 1
spread points = chart anchor * (basket return - chart return)
fair value    = chart close + smoothed spread points

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2.2 Residual Spread and Z-score

The spread measures how far the chart is away from the basket-implied fair value. The z-score converts that spread into a standardized reading relative to its recent history. This allows the indicator to distinguish normal noise from unusually wide relative-value deviations.

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z-score = (spread points - rolling spread mean) / rolling spread standard deviation

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2.3 Dislocation Is Not a Trade

A dislocation means the chart is far from basket fair value. It does not automatically mean the gap will close. In squeeze, earnings, halt-risk, borrow-stress, or news regimes, the spread can continue to widen. The regime-gated version therefore requires both statistical validity and price/spread confirmation before printing ARB+ or ARB-.

3. Tutorial: Reading the Overlay

3.1 Synthetic Fair Value Line

The main overlay is the Synthetic Fair Value line. It is plotted directly on the price chart so you can compare the current chart price against the basket-implied level.

Color / behavior	Meaning	How to use it
Orange line	Neutral or non-extreme spread.	No standalone signal. Use as context.
Lime line / green state	Chart is below synthetic fair value. The chart is relatively cheap versus basket.	Watch for D+ first, then wait for gates and compression before ARB+.
Red line / red state	Chart is above synthetic fair value. The chart is relatively rich versus basket.	Watch for D- first, then wait for gates and compression before ARB-.
Line far away from price	Large spread in price points.	Useful only if basket relationship remains valid.
Line drifting with price	Basket is moving with the chart.	Usually a healthier relative-value environment.

3.2 Background Shading

If background shading is enabled, the chart background highlights extreme dislocations. Green shading means the chart is below basket fair value. Red shading means the chart is above basket fair value. Treat shading as an alert zone, not an entry zone.

3.3 D+ and D- Markers

Marker	Meaning	Action
D+	Step 1 long-side dislocation. Chart is below fair value.	Observe. Do not treat as a confirmed long signal.
D-	Step 1 short-side dislocation. Chart is above fair value.	Observe. Do not treat as a confirmed short signal.

These markers are intentionally separate from ARB+ / ARB-. They show where the spread is wide enough to care about, but they do not confirm mean reversion.

3.4 BLOCK+ and BLOCK- Markers

Blocked markers are optional debug markers. They show when a dislocation exists but at least one gate prevents the confirmed signal. Turn them on while tuning the indicator or studying a false-signal environment.

Blocked reason	What it usually means
Basket valid = FAIL	Correlation, R2, beta, or active symbol count is insufficient. Fair value is unreliable.
Event/squeeze = KILL	RVOL, ATR, gap, residual jump, manual kill, or extreme

	widening suggests a flow-dominant regime.
Compression = FAIL	Spread is not yet closing. Cheap can keep getting cheaper; rich can keep getting richer.
VWAP confirm = FAIL	If enabled, price has not reclaimed VWAP for ARB+ or rejected below VWAP for ARB-.

3.5 ARB+ and ARB- Labels

These are the confirmed signals. A confirmed label means all active gates passed at that bar. It still does not guarantee a profitable trade; it means the indicator has upgraded the condition from dislocation to confirmed relative-value setup.

Label	Directional bias	Required behavior
ARB+	Long bias / chart cheap versus basket	Chart below fair value, basket valid, event clear, residual z-score compressing downward toward zero, optional VWAP reclaim passed.
ARB-	Short bias / chart rich versus basket	Chart above fair value, basket valid, event clear, residual z-score compressing upward toward zero, optional VWAP rejection passed.

3.6 Regime Gate Dashboard

The table in the upper-right is the operational cockpit. It should be read from top to bottom before accepting a signal.

Dashboard row	Meaning	Preferred reading
ARB Regime Gate	Current state: neutral, dislocation, blocked, or confirmed.	Confirmed ARB+ or ARB- only when Step 5 prints.
1 Dislocation	Whether z-score is beyond the configured threshold.	YES begins the evaluation. NO means no setup.
2 Basket valid	Whether relationship metrics pass.	PASS is required. FAIL means fair value is informational only.
3 Event/squeeze	Whether event/squeeze risk is clear.	CLEAR required. KILL means do not fade the move.
4 Compression	Whether residual spread is moving back toward zero.	PASS required. FAIL means wait.
5 Signal	Final gate.	PRINT means ARB+ / ARB- can appear. WAIT means no confirmed signal.
Spread pts / %	Distance between current price and synthetic fair value.	Larger is more interesting, but only if gates pass.
Z-score	Standardized spread.	Positive = chart below FV; negative = chart above FV.
Corr / R2 / Beta	Relationship health metrics.	Higher corr/R2 and reasonable beta are preferred.
RVOL / ATR exp	Event stress metrics.	High values increase risk of forced-flow regimes.

Z slope	Direction of residual change.	For ARB+, slope should be negative. For ARB-, slope should be positive.
Active symbols	How many basket symbols are contributing.	Should meet or exceed minimum active symbols.

3.7 Alerts

Alert	What it means	How to use
ARB+ confirmed	All long-side gates passed.	Candidate long-bias relative-value setup.
ARB- confirmed	All short-side gates passed.	Candidate short-bias relative-value setup.
Basket dislocation only	Step 1 only.	Useful for watchlist scanning, not direct entry.
ARB mean reversion	Spread crossed back through zero/mean area.	Useful as a profit-taking or review marker.

4. Settings Guide

How to tune the indicator without turning it into a curve-fit machine.

4.1 General Settings

Setting	What it controls	Raise / increase when	Lower / decrease when
Basket timeframe	Timeframe used to pull basket prices.	You want smoother basket reads, e.g. 5m or 15m while viewing noisy 1m charts.	You want exact chart-timeframe sensitivity.
Return anchor	Compares returns from previous close or session open.	Use Prev Close for full-session relative value and overnight context.	Use Session Open to isolate intraday leadership/laggard behavior.
Weighting	Auto Market Cap or Manual weights.	Use Auto Market Cap for broad mega-cap baskets.	Use Manual for custom peer groups, equal weights, or sector-specific baskets.
Spread EMA length	Smoothing applied to spread points.	Signals are too noisy or flickery.	Signals lag too much.
Z-score lookback	History used to standardize spread.	You want stable thresholds and fewer false extremes.	You want faster adaptation to current volatility.
Dislocation threshold	Minimum z-score for D+ / D-.	Too many low-quality dislocations.	Too few alerts or you are scanning very quiet pairs.
Minimum active symbols	Required number of valid basket contributors.	You want stronger data quality.	You are using a small custom basket.

4.2 Basket Symbol Settings

The basket is the most important part of the model. The indicator can only be as good as the peer group. A basket should share drivers with the chart symbol: sector, factor exposure, index membership, macro sensitivity, beta profile, and liquidity regime.

- **Use true peers first.** For a rental-car stock, use travel, rental, auto services, consumer discretionary, or index/sector ETFs rather than unrelated mega-cap tech names.
- **Avoid single-name event contamination.** If a peer has earnings, litigation, merger news, or a squeeze of its own, disable it temporarily.
- **Manual weights are useful.** Use them when one or two peers dominate Auto Market Cap weighting but do not actually represent the chart symbol.
- **Minimum active symbols protects the model.** If too few symbols are active, the synthetic fair value becomes fragile.

4.3 Step 2: Basket Relationship Validity

Setting	Purpose	Typical range	Interpretation
Relationship lookback	Bars used to calculate corr, R2, and beta.	60 to 200	Shorter reacts faster; longer is more stable.
Minimum correlation	Requires chart and basket to move together.	0.20 to 0.50	Higher = cleaner relationship; lower = more signals but weaker anchor.
Minimum R2	Requires enough variance explained by basket.	0.05 to 0.25	Higher blocks more broken relationships.
Minimum beta	Blocks near-zero beta relationships.	0.05+	If beta is too close to zero, basket has little explanatory power.
Maximum beta	Blocks unstable or explosive beta relationships.	3 to 6	Very high beta often means the chart is in a different regime.

4.4 Step 3: Squeeze / Event Regime

The event gate is designed to stop the model from fading moves that are dominated by forced flow rather than normal relative-value behavior.

Setting	What it detects	Tighten when	Loosen when
Manual event/squeeze kill	Your discretionary override.	News, halt risk, earnings, squeeze headlines, borrow stress.	Normal trading conditions resume.
RVOL kill	Unusual volume.	Studying squeeze-prone names or thin floats.	The symbol always trades high relative volume.
Bar move ATR kill	Large intrabar move.	You want to avoid momentum air pockets.	You accept more trend/reversal risk.
Gap ATR kill	Large gap from prior close.	Earnings/news days.	The market has absorbed the gap and stabilized.
ATR expansion kill	Volatility regime expansion.	False signals happen during volatility bursts.	You want more entries in active names.
Z-score jump kill	Sudden spread shock.	Spread spikes are creating traps.	Spread changes are too slow to trigger.
Extreme z-score kill	Extreme residual still widening.	You want to avoid CAR-style cheap-getting-cheaper setups.	You have external evidence squeeze risk is gone.

4.5 Step 4: Residual Compression

Setting	Purpose	More conservative	More aggressive
Residual slope length	Measures z-score trend.	Longer length, e.g. 10 to 14.	Shorter length, e.g. 3 to 6.
Compression confirmation bars	Requires spread to improve over N bars.	Higher value, e.g. 5 to 8.	Lower value, e.g. 1 to 3.
Minimum z-score compression slope	Requires a meaningful turn.	Higher slope threshold.	Lower slope threshold.
VWAP confirmation	Requires price reclaim/reject relative to VWAP.	Enable it.	Disable it.
Signal cooldown bars	Prevents repeated labels.	Increase to 30 to 60.	Decrease to 5 to 15.

5. Signal Playbook

5.1 The Five-Gate Checklist

Gate	Question	Pass condition	Fail behavior
1 Dislocation	Is the chart far enough from basket fair value?	D+ or D- appears; z-score beyond threshold.	No setup.
2 Basket valid	Is the basket still explaining the chart?	Corr, R2, beta, and active symbols pass.	Treat fair value as unreliable context only.
3 Event clear	Is price discovery normal enough to fade?	No manual kill, no RVOL/ATR/gap/z-shock kill.	Do not fade. Wait for regime normalization.
4 Compression	Has the residual started closing?	Z-score moves toward zero and absolute z shrinks.	Wait. Dislocation is still widening.
5 Signal	Are all gates aligned after cooldown?	ARB+ or ARB- prints.	No confirmed signal.

5.2 What a High-Quality ARB+ Looks Like

- The chart is below synthetic fair value, but not in a fresh one-way collapse.
- D+ appeared earlier, then the spread stopped expanding.
- Basket Valid shows PASS. Correlation is positive, R2 is acceptable, and beta is not extreme.
- Event/squeeze shows CLEAR. RVOL and ATR expansion are not screaming forced-flow regime.
- Compression shows PASS. Z-score is moving down toward zero and absolute z is shrinking.
- Optional VWAP confirmation has passed: price reclaimed VWAP or is holding above it after reclaim.
- The trade has a clear target: fair value line, z-score near zero, VWAP, or prior liquidity level.
- The trade has a clear invalidation: new residual expansion, new low after signal, event gate flips to KILL, or Basket Valid fails.

5.3 What a High-Quality ARB- Looks Like

- The chart is above synthetic fair value, but not in a fresh upside squeeze.
- D- appeared earlier, then the spread stopped expanding.
- Basket Valid shows PASS and beta is reasonable.
- Event/squeeze shows CLEAR. No abnormal upside volume/ATR shock is dominating.
- Compression shows PASS. Z-score is moving up toward zero from a negative extreme.

- Optional VWAP confirmation has passed: price rejected VWAP or is holding below it after rejection.
- The target and invalidation are pre-defined before entry.

5.4 Bad Signal Conditions to Avoid

Condition	Why it is dangerous	What to do
D+ keeps printing while price stair-steps lower	The model is detecting cheapness, not reversion.	Wait for compression or disable signal with stricter event filters.
Basket Valid = FAIL	Fair value anchor is broken.	Do not trade the spread.
Event/squeeze = KILL	Forced flow may dominate.	Avoid fading; reassess after volatility cools.
Z-score still widening	Spread is moving against the mean-reversion idea.	No ARB+ / ARB- until Step 4 passes.
Too few active symbols	Synthetic return may be unstable.	Add/repair basket symbols or lower minActive only intentionally.
Peer earnings or single-name news	Basket return is contaminated.	Disable affected peer or switch to ETF basket temporarily.

5.5 CAR-Style Failure Mode

A CAR-style squeeze/unwind can make the old version print repeated ARB+ labels because price is below fair value. In that regime, the right read is not "cheap, buy." The right read is "fair value is temporarily unreliable until flow normalizes and residual compression confirms."

CAR-style rule

When a stock is moving because of short-covering, borrow stress, option/dealer hedging, halt risk, or forced liquidation, the basket fair value line becomes a warning label, not an entry trigger.

6. Whitepaper Methodology

6.1 Abstract

This indicator implements a relative-value framework for intraday or multi-timeframe chart analysis. It estimates synthetic fair value from a selected peer basket, measures residual deviation with a z-score, then applies regime gates to distinguish statistical dislocation from confirmed mean-reversion setup.

6.2 Model Construction

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For each active basket symbol i:
  peer_return_i = peer_close_i / peer_anchor_i - 1
  weight_i      = auto market cap weight or manual weight

Synthetic return:
  synthetic_return = sum(weight_i * peer_return_i) / sum(weight_i)

Chart return:
  chart_return = chart_close / chart_anchor - 1

Spread points:
  spread_points = chart_anchor * (synthetic_return - chart_return)

Synthetic fair value:

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fair_value = chart_close + EMA(spread_points, smoothLen)
```

6.3 Signal Model

Dislocation:

ARB+ side when zScore > zThresh
ARB- side when zScore < -zThresh

Relationship validity:

correlation >= minCorr
R2 >= minR2
beta between minBeta and maxBeta
active symbols >= minActive

Event/squeeze block:

manual kill OR high RVOL + large ATR move OR large gap OR ATR expansion OR z-score jump OR extreme residual still widening

Compression:

ARB+ requires z-score slope down toward zero and absolute z shrinking
ARB- requires z-score slope up toward zero and absolute z shrinking

Final signal:

confirmed signal = dislocation + relationship valid + event clear + compression + optional VWAP confirmation + cooldown OK

6.4 Why the Model Uses Gates

A spread can be wide for two very different reasons: temporary relative-value mispricing or regime break. A naive mean-reversion signal cannot reliably separate these states. The regime-gated model therefore uses a layered decision process. The first layer detects opportunity; the later layers try to filter out conditions where the opportunity is not yet tradeable.

Layer	Purpose	Risk reduced
Dislocation layer	Finds unusual spread.	Missing potential relative-value opportunities.
Relationship layer	Confirms the basket still explains the chart.	Trading a broken or irrelevant fair value anchor.
Event layer	Blocks abnormal flow environments.	Fading squeezes, halts, earnings shocks, and forced liquidation.
Compression layer	Waits for spread improvement.	Catching falling knives or shorting squeezes too early.
Cooldown layer	Limits repeated labels.	Overtrading the same spread impulse.

6.5 Assumptions and Limitations

- The model is not true arbitrage. It is a synthetic relative-value approximation.
- The model does not directly ingest short interest, borrow cost, options open interest, gamma exposure, halts, or real-time news.

- Auto Market Cap weighting depends on available financial data and may not update the way a live institutional basket would.
- Correlation and beta are backward-looking. They can look valid until a regime break happens.
- Low-liquidity or thin-float names can invalidate basket relationships quickly.
- A confirmed ARB+ / ARB- signal is a research signal, not a standalone order instruction.

6.6 Validation Framework

The correct way to improve the indicator is not to change settings after every losing trade. Instead, label a sample of signals and review performance by regime.

Metric	Question answered
Hit rate by gate state	Do signals work better when all gates pass versus dislocation only?
Time to mean reversion	How many bars does it take for z-score to approach zero?
Maximum adverse excursion	How far does the trade move against the signal before improving?
Maximum favorable excursion	How much opportunity appears after signal?
Outcome by RVOL/ATR regime	Do high-event regimes need tighter kill settings?
Outcome by basket validity	Which corr/R2/beta thresholds improve quality without eliminating too many signals?

7. Calibration Presets

Use these as starting templates. Do not treat them as universal values. Each symbol, timeframe, and basket should be calibrated with actual forward review.

Preset	Best for	Suggested changes
Balanced	Default watchlist monitoring.	zThresh 1.5, minCorr 0.25, minR2 0.10, RVOL kill 4.0, ATR expansion kill 2.0, VWAP confirm off, cooldown 20.
Conservative	Higher-quality but fewer signals.	zThresh 2.0, minCorr 0.40, minR2 0.20, RVOL kill 3.0, bar move kill 1.7, VWAP confirm on, cooldown 30.
Aggressive	Fast scanning and early warning.	zThresh 1.2, minCorr 0.15, minR2 0.05, RVOL kill 5.0, VWAP confirm off, cooldown 10.
Squeeze-prone	Names with short-interest, borrow, meme, or halt risk.	zThresh 2.0, minCorr 0.35, minR2 0.15, RVOL kill 2.5, gap kill 1.0, extremeZKill 3.0, VWAP confirm on, manual kill during headlines.
ETF / index	Broad liquid ETFs or index futures proxy charts.	zThresh 1.3 to 1.7, minCorr 0.40+, minR2 0.20+, RVOL kill can be looser if liquidity is deep.

8. Trade Review and Risk Controls

8.1 Entry Checklist

- Which side is the setup: ARB+ or ARB-?
- Is the basket relevant to the chart symbol today?
- Did Step 1 identify a dislocation, or are you forcing a trade in neutral state?
- Does Basket Valid show PASS?
- Does Event/Squeeze show CLEAR?
- Does Compression show PASS?
- Is VWAP confirmation enabled and satisfied, if required?
- Where is the target: fair value line, z-score mean, VWAP, or liquidity level?
- Where is the invalidation: spread re-expansion, new extreme, gate failure, or price structure break?

8.2 Invalidation Rules

Invalidation	Interpretation
Event/squeeze flips to KILL	Flow regime changed; the setup is no longer normal relative value.
Basket Valid flips to FAIL	The fair value anchor is no longer trustworthy.
Z-score expands beyond signal bar extreme	Mean reversion failed or is too early.
Price makes a new trend extreme after signal	Market structure is not confirming the fade.
Peer basket becomes contaminated	A basket component has its own event risk.

8.3 Exit / Target Ideas

- Partial target at VWAP if the trade is intraday and VWAP was the confirmation level.
- Primary target near synthetic fair value or when spread points shrink materially.
- Statistical target when z-score approaches zero or crosses the mean.
- Time stop if residual does not continue compressing after a defined number of bars.
- Exit immediately if the event gate flips to KILL after entry.

8.4 Journal Template

Field	Record
Symbol / timeframe	Ticker and chart timeframe.
Basket symbols	Active peers and weighting mode.
Signal type	D+ only, D-, ARB+, ARB-, blocked attempt.
Z-score at signal	Residual extremity.
Corr / R2 / beta	Relationship quality at signal.
RVOL / ATR expansion	Event stress at signal.
Compression state	Z slope and absolute z behavior.
Entry / stop / target	Execution plan.
Outcome	MFE, MAE, time to target, time to invalidation.
Notes	News, options flow, earnings, borrow, macro conditions.

Appendix A: Glossary

Term	Definition
ARB+	Confirmed long-bias relative-value signal: chart cheap versus basket and gates passed.
ARB-	Confirmed short-bias relative-value signal: chart rich versus basket and gates passed.
D+ / D-	Dislocation-only marker. Step 1 passed, but no confirmed signal yet.
Synthetic fair value	Chart-level estimate based on weighted peer returns.
Spread points	Point distance between chart and synthetic fair value.
Z-score	Standardized spread relative to recent history.
Residual compression	Spread moving back toward its mean / z-score moving toward zero.
Basket validity	Relationship health using correlation, R2, beta, and active symbols.
Event/squeeze kill	Filter for abnormal flow regimes that can break mean reversion.
RVOL	Current volume divided by rolling average volume.
ATR expansion	Current ATR percent divided by its baseline.

Appendix B: Quick Reference Card

Observation	Read
Orange fair value line	Neutral context.
Green shading or D+	Chart below fair value. Watchlist alert only.
Red shading or D-	Chart above fair value. Watchlist alert only.
Basket Valid = FAIL	Do not trade the fair value signal.
Event/Squeeze = KILL	Do not fade the move.
Compression = FAIL	Wait. Spread is not yet closing.
ARB+ prints	Confirmed long-bias relative-value setup.
ARB- prints	Confirmed short-bias relative-value setup.
Z-score returns to zero	Mean reversion target / review point.

Appendix C: Common Failure Modes

Failure mode	Symptom	Fix
Wrong basket	Low corr/R2 or signals that do not behave.	Use true peers or sector ETFs; disable irrelevant symbols.
Too sensitive	Constant D+ / D- markers.	Raise zThresh, increase zLen, increase smoothLen.
Too slow	Signals appear after move is mostly	Lower smoothLen, lower confirmBars,

	over.	shorten relationship lookback.
Squeeze traps	Repeated dislocations while price trends one way.	Tighten event filters, enable VWAP confirmation, use manual kill.
Overtrading	Multiple labels in same impulse.	Increase cooldown bars and use blocked markers for debugging.
Contaminated peers	Basket jumps for unrelated reasons.	Disable peers with news or earnings; use ETF replacement.

Bottom line

The indicator should be read as a regime filter plus relative-value signal. The fair value line finds the spread. The gates decide whether the spread is tradeable. ARB+ is not cheapness; ARB+ is cheapness plus a valid relationship, clean regime, and confirmed residual compression.